CCN 2019: SE-CC: Challenges and Controversies: The Free Energy Principle

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okay let's start good afternoon everyone it's my pleasure to welcome you to cc ends challenges and controversies session where we will collectively discuss and hopefully debate the free-energy principle so as you may know the free-energy principle or act of inference has been developed and applied in the neuroscience community for the past 15 years particularly in the field of cognitive neuroscience and psychiatric research and somewhat in parallel it's developed into a good model and robust debate amongst philosophers particularly there in the field of consciousness research and embodied cognition and even more recently the framework has reached formally into physics where the maths have been unfurled to reveal Schrodinger's equations Maxwell's law of electromagnetism and wave particle duality and this is all afforded by dynamical Systems Theory and Markov blankets where spatio-temporal recursions of a single idea the drive to minimize free energy speak speaks to thickness across multiple scales from quantum phenomena to large active particles like ourselves so today in this session I hope we can cover the core concepts and mathematics and its implications for the study of the human brain and cognition and the format for the next hour and a half we'll see introductory remarks from our discussants for 15 minutes or 20 minutes apiece followed by hopefully a lively debate and Q&A from the audience so there'll be no provocation here of our Parliament and to facilitate audience participation we'll have one roving mic and I've also set up a poll everywhere channel where you can submit your questions live now ordering the introductions or during the discussion via your smartphone or your laptops and so to do this you go to poll evey calm and enter a user name which is me Rosalyn M 0 92 or you can go directly to poll evey comm forward slash Rosalyn M 0 92 and it's written on the board's here and there you can submit your questions and I'll pass them to discussants depending on how much time we have we'll get through as many as we can if we have a lot of questions I'll be updating the live pose so be patient if the poles fill up be putting a new pole on the same address and do add your name and institution to the end of the question if you so wish okay - so to our discussants they say that a good boxing match is not based on opponents of different abilities but on matching opponents of different styles and so today I think we'll have a very good boxing match since in the red corner and in the blue corner we have two exceptional scientists who have in some ways sympathetic views and ideas but distinctive points of view so in the red corner we have professor Christine Karl is the scientific director of the welcome to a center for human ear imaging at University College London Karl is the originator of the free energy principle and you might say has published extensively in the theory from its development to applications across the fields I've mentioned and in the blue corner we have professor Jeff Beck Jeff is professor of Neurobiology and biomedical engineering at Duke University and at Dukes Institute of brain sciences and he is well known to this community for his seminal contributions to probabilistic coding bayes in the brain and sensory motor domains and also in higher cognitive functions studying decision-making across species so thank you both sincerely for agreeing to come and participate so from the rumble in the jungle to the Thrilla in Manilla we now have the Battle of Bayes in Berlin and so with that for round 1 let's welcome card first open [Applause] well thank you for our invigorating introduction and it's clearly a great pleasure to be here and I'm flattered in a perverse sort of way you have to be controversial I don't think I've normally cast as controversial though so that's good and the last time I had to do something that was controversial was to discuss different ideological positions under the banner of NYX versus Scruffy's I don't know if you if this is a sort of an dichotomy which people are aware of I liken this to a sketch a sort of pre Monty Python sketch in England where class wars were the thing in the 1970s and 80s and here we have John Cleese and the Two Ronnies talking about the class that they come from and this tall gentleman here is upper-class and he looks down upon this gentleman here who is middle-class who in turn looks down upon Ronnie Corbett who in this sketch is very lower class and I'm gonna make the position just to situate the debate I'm gonna make the argument that this hierarchy of ideological high church nuts is paralleled in the NEETs versus the Scruffy's and the Mystics from our point of view I'm going to associate the needs with physicists people who are in search of simple unifying neat explanations for everything and I want to contrast that with engineers who are the Scruffy's and these are people who are much more pragmatic and they want to know what works so they will adopt anything that works and from our perspective that could be 20th century behaviorism reinforcement learning drift diffusion models temporal discounting if it works use it and then we have the Mystics now I usually have psychologists under that but I thought the you got to keep the crowd on the side Avenue and so I've made them I've made them philosophers we don't need to talk about them any further so in four slides my brief is false lies leave the free energy principle so here is the free energy principle in four slides at its simplest it just says everything that can change will change to minimize very variation or free energy and that's sometimes articulated written in philosophy as self evidencing now free energy here I've expressed in terms of complexity - accuracy and that mathematically provides an upper bound on the quantity known as log evidence or disc evidence or marginal likelihood that's the probability of some outcomes or observations given a model or my understanding of how those those data were generated so I'm going to focus on this decomposition as a way of understanding the role of minimizing complexity whilst providing an accurate explanation for the sensory observations at hand and motivate the importance of that just by unpacking the different kinds of interpretation one can bring to bear to understand what this quantity is here the log probability of some outcomes at some time possibly in the future given a model of how those outcomes were generated and if you associate that log probability with value what we're basically saying is that minimizing free energy is the same thing as minimizing a bound on value and I should say if you come from machine learning this is exactly the same as the elbow or the evidence lower bound using things like variational auto-encoders but from point of view or psychologists they could also be value they could be the sort of expected reward that you will obtain the a particular course of action so this can be regarded as a statement of the imperatives that underwrite reinforcement learning optimal control theory and indeed in economics expected utility theory so that's nice that covers a large bunch of people there are other groups of people who would think more in terms of information theory so the negative of this value is called self information in information theory also called surprisal and it is the quantity that's upper bounded by the variation free energy and minimizing self information or minimizing surprisal is essentially the same as maximizing the mutual information and from that you can spin off the principles of minimum redundancy the principles of maximum efficiency the infamous principle of ralph olinska and or and indeed the free energy principle itself that in turn is nice because the expected the time averaged of self information is known as entropy so what we're saying is if everything has to minimize its free energy then it has to basically minimize over time entropy and of course that is the holy grail of self-organization in physics synergetics of course if you're a physiologist it's just homeostasis is just keeping outcomes within physiological bounds preventing the dispersion and the tendency to disorder that one normally attributes to the second law of thermodynamics but there's a final interpretation here which might appeal to many of you in the audience and that's is drilling down on this marginal likelihood itself so if I interpret M as a model of a journeyer model of how these data were generated then this becomes model evidence also known as integrated evidence which means that minimizing free energy is simply a statement of the fact that or an imperative to maximize the evidence for one's models of the world and from that we can develop the basic brain hypothesis evidence accumulation and things like predictive coding and that's largely the perspective that I'm going to adopt on free energy minimization the rather move entailed by the free energy principle which is probably less familiar to too many people it's not just about inferring the best explanation for the causes of your sensorial it's also inferring what you're going to do it's also about what are the best data or sensory samples that with my active palpation of the world I can gather together in the aid of minimizing free energy or maximizing the evidence for my own models and we're appealing here to very old ideas first elaborated by people like Lindley in terms of optimal Bayes in design so it's not about building good models of data its what are the best data you can go an actively sample from your world in order to afford the best kind of inferences about the causes and in that world and that's formally can be articulated in terms of perception being in the game of minimizing free energy basically the difference between your posterior beliefs about hidden or latent states of the world at some time possibly in the future in relation to your true posterior which is the bound and this is the log evidence here so policy selection in this expression here where PI denotes a policy is all about minimizing the free energy consequent upon taking a particular policy or acting on the world in a particular way and all we need to do is to take the average free energy under our beliefs our posterior predictive beliefs about the outcomes that would ensue if I did that and if we apply that to the free energy expression here we're now conditioning on a particular policy and we're just slightly relabeling and rearranging the terms to express it in terms of a sure of incentives and vow and values and I've written them down here in terms there are a number of ways of unpacking this and I'll do that in a second in terms of instrumental and epistemic value so what does that mean well if I just focus on components of the terms of the expected free energy and pretend that we had a an agent or somebody who had no particular prior preferences no value extrinsic value or reward preferences then we're left with these two terms here and it turns out that these are exactly or this is exactly the quantity that's optimized an optional Bayesian design and the visual neurosciences it's known as Bayesian surprise popularized I believe that Christophe Kok an idiom balding and mathematically that is exactly the same as a mutual information between outcomes and their causes in the future and of course that is consistent with the principle of maximum efficiency of minimum redundancy articulated by people like Horace Barlow let's just now take two sorts of uncertainty out of the mix and see what this expected free energy looks like I'm hoping that you will start to recognize the terms so the first sort of uncertainty is basically ambiguity I have no ambiguity about what's out there beyond my sensations I can see the states of the world directly and that just leaves us with these two terms here they're basically score the difference between what I will predict will happen and what I prefer will happen and that's known as a KL objective function for KL control in optimal control theory or if in economists it's risk sensitive control to minimize the outcomes of behavior in terms of my expected outcomes in relation what to talk to my goals or what I prefer to happen now I'm going to take the final source of uncertainty which is what will happen if I do do that and if we take both source uncertainty out we're right back to reinforcement learning expected utility theory so the expected outcomes on my prior preferences under what I think will happen the point here is that these are all internally consistent and equally valid ways of trying to write down often behavior but they are all special cases of one imperative which is to minimize the expected free energy or maximize expected evidence for your models of the world in the future I won't go into this in detail Roslin's is intimated you can extend this argument right down to quantum physics or right up to evolution and many people have started to do that by putting a Bayesian gloss on in theoretical biology in terms of evolution practically though I just want to conclude with a comment upon the explanatory scope and what this actually means for people creating artifacts or creating prediction machines to explain say fMRI responses in the ventral stream it is neat and simple that's the whole point about being a NEET or a physicist it's just basically saying what we have to do is to provide accurate accounts of our world or the sensed world that is as minimally complex as possible thereby maximizing the evidence for me as a model of my world and this complexity term is extremely important it's just Occam's principle and from that we can motivate certain factorizations that simplify a giant in model of the world you can get differential specializations in your anatomical functional anatomy that belies or speaks to that factorization we can talk about redundancy minimization the shape of receptive fields that maximize the information gathered from the environment efficient coding and so on we can also think about this from the point of view of pragmatism and economics what we're talking about because we're using an evidence band we talk about bounded rationality slightly disingenuous use of the word bounded rationality from the point of view in the columnist but it's still bounded rationality it's perfectly Bayes optimal but it's using an evidence band and that is the definition of approximate Bayesian inference based upon free energy bounds it also very gracefully accommodates heuristics that emerged in many different levels so in economics it would be bigger ends and like heuristics these are just priors empirical prize that have inherited from your parents or neurodevelopment or some other form of Bayesian model selection and if you're an engineer or a physiologist then via the joinin scale equality and Landau's principle is a direct relationship between your ability to minimize a complexity cost and the algorithmic complexity that underwrites the thermodynamic cost of erasing bits so if you've got the three energy minimizing solution you have a bound upon the most efficient way to do it whether you are dealing with the cerebral blood flow as a physiologist or whether you're actually designing supercomputers and you want to minimize the amount of electricity required to solve a particular computation so that's about it neat and simple evidence is equal to simplicity plus accuracy as Einstein said everything should be made as simple as possible but not simpler so there you have it thank you very much indeed [Music] Thank You Carl next up we have no sabotages so hopefully I see we're getting questions through this is great keep them coming so just to remind you all it's pole Evie comm forward-slash Rosalyn M 0 9 - well Jeff is getting ready and has miked up to respond to car sweet yeah we'll see so no thank you I have a word of advice for anyone who's ever invited to do something like this in the future read the fine print on the invitation because I got this nice little email and it said would you like to debate Carl first inand I didn't really read the whole thing I just said to myself that sounds like a ton of fun like what could be better but I was under the impression that we weren't talking about the free energy principle as a unifying concept for a cognitive computation but rather as a theory which has created a bit of a problem for me because as as a unifying concept for cognitive computation Carl hiring I think complete agreement this is to be an incredibly boring debate so I'm sorry maybe we'll throw something but it it's not gonna get too much better than that because we are as far as I can tell largely in agreement on this question there's another question the question I wished we were addressing which is is free energy principle unifying theory of cognitive computation and I would say I have a four type of very slightly slightly different answer very different answer which is somewhere between no and not yet or it needs more you know it there's it's it's it's not a complete theory and I'll try to get I'll try to explain what I mean by that at the very end but because we're largely in agreement on a lot of things first thing I'm going to spend a lot of time doing is addressing popular criticisms of the frit what I'm going to call the free energy framework instead of principle because that makes more sense and I'm not gonna refer to as a theory though neither did Carl oh I noticed I was tallying not a single use of the word theory saw it coming oh sorry I'm gonna put this down so I'm gonna spend a lot of time just talking about what are the common criticisms of the the free-energy framework and addressing them because I don't think they're valid criticisms I mean and we'll talk about why I don't think it's quite a theory or may not be a theory but it's nonetheless incredibly useful okay so here we go so here are the common criticisms of the free-energy framework they largely center around they don't center around things like epistemic and intrinsic value those words don't pop up very often instead they they tend to take the form of something more like wait a minute why it why this obsession about surprise after all there are very desirable things that are very surprising so if I'm minimizing surprise aren't I eliminating all of these desirable things like winning the lottery the second criticism comes with oh and you know there's actually really trivial way to minimize surprise you it's known as the darkroom problem what do you do if you want to minimize surprise while you just stop sensing the world cover your eyes go into a dark room and this has been a these Chris's have been addressed many times I think most in that's what was the name of the article that was the conversation between the philosopher and the physicist absolutely lovely I had trouble reading it but I'm gonna rephrase that as I go through and I think like maybe planar terms I hope but here's here's a little exchange that made me chuckle this is one of Karl's responses in a paper that he wrote years a few years back addressing this sort of issue regarding the active inference part of the free energy principle so he did something that made me chuckle I thought it was hilarious and brilliant in his actual paper he quoted a reviewer and then directly responded to him this was brilliant for two reasons so first of all what better way to placate review or two especially reviewer who'd use a word like a lid than by actually acknowledging his use of that word publicly completely brilliant of course the my second response to seeing this was I had to look up what the word lid was it turns out it's actually pronounced alive and what it means is to conflate so 10 points for saying for knowing what the word ye lid means and -5 for the missed opportunity for alliteration catastrophic ly conflating is a lot better way to say that and here's calls response I'm not gonna read Karl's response you can read it for yourself but I'm simply going to try to explain it with a few examples ok so here's here's how this goes oh sorry there was a actually what's Karl's response I will I will not read it I will simply summarize it was to say that that optimal behaved there are two kinds of surprise that you have to worry about when minimizing for energy an action selection in an environment which are also selecting your action one kind of surprise is about the sensory input and another kind of surprise is about your action and it and/or is about yet is about how surprising transitions between states become and that's what I'm gonna try Oulu say so here's here's a here's the basic problem as a lot of people see it in my opinion so this is a typical RL type problem here's a mouse he's an amazing ever been in this maze before he has a massive amount of state uncertainty because he didn't understand the configuration of the maze he doesn't know where the cheese is he doesn't know where the walls are there's a massive amount of uncertainty what does the mouse do he begins by exploring his environment so he zips around eventually he finds the cheese and in an RL framework he gets ten points he got the cheese it's over you pick him up you put him in another spot in the maze and then what is the mouse do well he now knows something about his environment so he set has he says oh I know something others environment now have an informed policy that will get me from point A to point B so I can get more cheese and therefore more points there's something that's missing from this framework and they're no I'll just wave my hands there's a step that's missing from this simple RL perspective which is which is yes there's a robot maxima doesn't we reward in the free energy framework and maybe that's a problem there's also something missing what does the mouse do when he gets to the cheese there's a serious question he eats the cheese right getting to the cheese yes it's rewarding and you can model that as reward but you can also model it as suddenly having a great deal of certainty about his actions he's not surprised to find cheese he's sure he's going to eat it I think that at a high level I'm hoping to get a nod and I'm not getting an OK at a high level I would say that that is the difference between the reinforcement learning and what I would call like the this free energy approach for an active inference it that there is this there's two notions of surprise that we're doing a sensory surprise and and surprise in your action space or surprise in your transition probability space matha apologize for the math but he started it his here's a simple in a simple Bayesian RL example I I'm sure that many of you are familiar with this that elucidates the difference between or how policy and reward are related so I'm going to frame this as a Bayesian reinforcement learning problem so initially you're a mouse you're in a maze you have some you know in the absence of a control signal which will eventually be you you have some sort of popu there are states that you would transition between maybe just moving randomly through your environment but of course you could exert a control signal and therefore U and therefore execute a particular policy what you'd like to do is you'd like to choose your control signal so as to minimize it's tempting it's so tempting so as to minimize your cost or maximize your reward they're the same thing how does that work well just for simplicity because we want a simple example with an explicit result we're going to assume that the cost of actions is simply measured in the KL divergence between your you know sort of go with the flow prior of just randomly moving through states as opposed to as opposed to exerting control as opposed observing the control signal U and then of course there is some cost or reward associated with getting the Chi or maybe just stagnating starving whatever and we're gonna call that C of s so that's like the cost of achieving the state that is I suddenly or the value of either not or achieving the state though in which you have cheese for example so these are logs we can do math we move things around and what we end up with is being able to show that our expected cost or our expected the negative of our expected reward can be expressed as a kale divergence between our policy can you see my cursor no of course not maybe I should be using this yes as the expected kale divergence between our policy and something that is kind of like a probability distribution not quite there's some constants that we're neglecting but they don't really matter what we do is we're gonna find the U that makes this as close to this as possible what should we call this right well okay we got it from this you know sort of single state cost it ends up looking something like this you can call that your policy right you can call that your policy prior right this is what you this is how you should adjust your control signal you want your behavior to be like this to maximize your reward or to minimize cost this is the endpoint of this kind of reward maximization calculation in contrast this is the starting point of a free energy framework you start with a policy prior I'm sorry I'm Zack pick house as I cannot use the word policy prior you start with a default policy which says how you would move from state to state that you like that is consistent with your goals but you've abstract it away reward entirely you just have this default policy so at the end of the day what is called really cog yeah we'll put it that what is called really done my view on this is what he's done is he's sort of come up with an equivalent but slightly different definition of what a decision-making organism is so here's the standard decision-making formulation right there's gonna be some data it's going to come in to your brain you're gonna do some inferences which is going to be something how you represent the world or represent your data and then generate some action so what do you need to do in standard decision-making for you to have a few things right an organism becomes defined by its inference engine how it interprets data from the world its defined by its reward function and explicit and implicitly the goal of maximizing reward okay this is the standard way to do it and of course what do you do with this information will decide you find a policy that maximizes your reward alternatively within the free-energy formulation or framework or whatever want to call it an organism is not defined by a reward but instead of defined by its inference engine which is an approximate distribution applied to a generative model this is an important caveat there doesn't have to be this is a little more general in the sense that doesn't have to be a generative model that informs the affected that informs the inference engine but there can be and a pal and a policy a default policy what you would do with you got to the cheese if you knew nothing about the environment or actually my brand we don't even know about the cheese if you do find the cheese your actions become very clear and that that sort of thing is incorporated into this sort of default policy the probability of eating the cheese given that you have cheese in front of you and then of course there's the aesthetically pleasing feature that if you take this approach beliefs and actions can be both selected by minimizing a single objective function which is called free energy it's pretty so before I move on I just like to pull the audience now not electronically that's really intimidating just hands what makes more sense like reward or policy as a way of another equivalent they're mathematically if you have a reward function you can find an equivalent policy that maximizes it and to some extent vice versa it's a little messy but it can be done but which is better right well it's it's my view that it's really hard to say I kind of have a sneaking suspicion that from an empirical perspective policy is a better thing to think about because policy is what we actually measure in the lab so here's how do you measure reward well you don't major directly you infer it you put somebody in a room when you said you want Snickers or skittles what do you do you behave you perform an action you say Snickers if you're a rational human being now you can change this a little by saying okay well okay skittles or 75 percent chance of Snickers and maybe get a better estimate of their reward but at the end of the day what you're actually measuring is people's behavior it's the actions that people are taking when placed in an unambiguous state that is a policy you always directly infer policies in an experimental setting and never infer reward directly right the reward is something you compute from observations of someone's policy so from an empirical perspective this actually makes a little bit more sense and so I like it just a couple more things I have to cover even though I think they're lame so there's the darkroom problem I think that some of these criticisms qualify as lazy or - well now I shouldn't have said that I take that back I think some of these credits some of the criticisms the phones of this category are too high level for me to understand so minimizing surprise is too easy there is a great way to minimize your sensory surprise you can go into a dark room you can cover your eyes you can simply stop participating in the world and that's all fine and dandy now there's the trivial response to this trick this crazy criticism is to say yeah that's not gonna last very long right if you if you close your eyes you will eventually die you will stop making predictions and everything will be surprising now there's a bit of a sticky wicket here because if your actual prior says with great likelihood you live in a dark room your predictions will be your prior predictions and they'll state still be pretty good now we can play this game forever turns out oh right so the response is well total surprise minimization actually encourages life you wouldn't do this okay so I call this the pit coward joinder this comes from my good friend Zack who's seated over there and is probably going to bury his head in a second when a paraphrase a little conversation we had yesterday I'm gonna change the words to make me look good and look bad but it goes something like this what if I brought nachos into my dark room where I'm blindfolded and my eyes are closed I'm not going to die you might need a few other things if you wish to survive but at the end of the day yeah that's what seems to be like a perfectly acceptable solution to the problem of minimizing surprise so what's your problem Zach what's wrong with this solution does that would say no exact would say I would never do that and I would say well I I mean it doesn't seem that bad bring an Xbox you know that actually sounds a lot like my personal time that said other organisms find this acceptable what the hell's wrong with you Zach right so an example of this is mold ah perfectly timed perfectly timed done to pushing buttons this is a perfectly acceptable solution to some organisms that's one of them believe it or not that's actually a picture of mold on nachos so what's the next problem the problem is is that his default policy is the policy of a human being and not of mold this becomes this is a perfectly acceptable solution to this he's not mold I'm pretty sure and so of course he's not going to pursue that as a solution but there are organisms that do because that's their policy policy has become something that now defines an organism instead of reward and that's really all I wanted to say so I hope that at this point there you've come to a few conclusions in a timely fashion I would say that reward functions and default policies are equally valid ways of defining decision making organism the free energy principle is nice in the sense that it provides a general and sensible mathematical framework for description of inference and decision making it's aesthetically pleasing and so on and so forth I hope you draw 1/3 conclusion with this this is the worst debate ever because I've just spent my 20 minutes talking about what's right about the free and principal or the free-energy framework well that's because I haven't gotten to the last man so it is the free-energy framework I insist on calling it unifying and I would argue that yes it is it is precisely because it doesn't make a lot of assumptions and most of the assumptions that it does make you can kind of squeeze other things into right so here the fundamental assumptions as I see it of the free-energy framework right the brain has a model that includes a policy it has a set of accessible approximations for the purposes of performing inference an action selection which we call cue right and inference an action selection occurs simply by minimizing the you know either klq P or free energy or whatever you want to call it as a result of it it's some incredibly general theory right anything can be in court on anything know anything I'm gonna stick with anything just about any description of a behaving entity can be squeezed into the free energy from by judicious choice of your model that you're inverting p.m. which is what I called it and your choice of approximations q anything and I spent a lot of time trying to think of things that you couldn't include in the free energy principle and I actually kind of initially I sort the oh well you know sampling based inference is a unique kind of inference me know that you can totally squeeze that into example what's a sample after all right what can you do with the sampling so you have 50 samples what can you do with them well I can evaluate free energy I can get a point estimate I can take an average and to the extent that my sampling scheme is a good one on average free energy will be going down using a sampling based difference game okay so that that gets in there too so we kind of leaned to get rid of the what is not how about non-bayesian schemes totally you can get non yes this is very busy in my definition there is a Bayesian break we do take into account uncertainty but maybe not always how can you fit a non-bayesian scheme it is well again by judicious toit choice of Q right typically we like Q because Q can contract variance and uncertainty about the properties of the world that we care about but why not pick a Q that doesn't why not pick a Q that just says oh zero entropy Q's that's all I'm allowed now you're basically doing map estimation and you're not behaving in a Bayesian way that's in there anyway so it can do anyway and that's why I called a framework and not a theory right it's great it's a great mathematical language it's a language that I hope we would all adopt because I understand it it makes sense to me but it provides very little guidance regarding the things that I actually care about right if there's no guidance within the pen I could be wrong there's no guidance within this framework for choosing your generative models that the brain is using there seems to be very little guidance for choosing q there is some I would not say there's none but there's very little guidance for how you to choose the kinds of approximations that the brain is using there's as far as I can tell zero guidance for the optimization algorithm and so here's a few that people like and more critically at least to me there's almost no guidance in terms of the relationship between your beliefs in this case your cues right and neural activity it's simply agnostic and that's what makes it a framework and not a theory I just wanted to quote two peoples who said very smart things and phrase things much better than I could Sam guess gee is Sam Sam Gershwin and he his quote is that predictive failures of the free energy he said principle I say framework are typically attributed to poor choice of your P's and your Q's if you don't mind your P's and Q's but you know you yes thank you I got one laughs out of that one thank you Eric if you don't mind your P's and Q's you will make bad predictions and if you do you can make good ones but that's not a property of free energy it's a property of your choice of models and an inference algorithms and Ralph Hefner is our H in this scenario he basically reader the same thing using much better words than I would have chosen that is that thank you Thank You Jeff let's leave those up shall we and Dimitri I'm gonna give you a microphone so you can wander around the crowd and look for questions and we have some online and maybe we'll take a few minutes for Carl to address some of the points yeah this is fine thank you that we just heard whichever you'd like yeah thank you well I thought you'd use all the good words usually in the right order the and I like that P&Q thing I never thought about before so for those of you don't know that comes from print in the old days when they used to print and these two have different blocks or P's and Q's and when apprentices used to get them inverted I also do prepositions and questions the like it's just actually literally is if he's getting the P's and Q's right so to take your in to take your lust I think you've identified a number of really key issues and unfortunately we're going to be in largely agreement here so you have framework I think is perfectly fine I use the word principle because it's essentially the free-energy principle is just a variational principle of least action you're formally mathematically that is what it is but if you want to use that to provide a framework within which to house or consider various process there is I think there would be a very appropriate use of the word of a framework but I slipped in there the distinction between a process a theory which may or may not be consistent with the free-energy principle and the framework and the principle itself and I think you're absolutely right that you know the the framework does not prescribe a particular process theory and inversion scheme and more crucially it doesn't provide what we all really want to know which is that the nature and the form the physiology of the implicit generative models to which you apply the variational machinery under the framework or the support and I think there's a implicit in some of your statements which I agree with entirely was only so here is decree cost to the complete class theorem or theorems that say for any pair of observed behaviors and loss function there is a Bayes optimal solution so I think you're absolutely right the free-energy principle cannot be wrong yeah it cannot be falsified of course in the sense of a variational principle of least action it doesn't have the attribute of being falsifiable but any process theory can because to be a process sir you'd have to write down the form of the Geritol model what sorts of quantities entails are you including inference about fast hidden states are including learning about slow model parameters are you including structure learning are including evolution so all of these start to shape the form and separation temple scales in the hierarchical form of the Geraldton model at hand and of course that's when all your processor has come in I use sampling schemes basing Kalman filtering predictive coding belief propagation variational message passing all of these are processed there is which you really have to you know commit to and I think at that point then the game is is basically what we all do with it in our daily lives you know there's no great gift there it's just having a framework that highlights the connections between them the other point you were making which I'd haven't thought about before was inverting the the way that one defines the right sort of behavior in terms of the outcomes and reward or cost functions and is looking at that as real way of writing down the kinds of things that I do you know there are two equivalent ways and I think formally that probably you could articulate that argument on the basis of the complete class theorem so it's probably provably true which means as you intimated II it's a question of personal choice which you which you commit to or you write about all the rhetoric that you use the world's one point though which I thought deserved a bit more attention you seem to have I was going to use the word Elijah's but I can't baby you'll certainly have alluded you have alluded the epistemic part it is not the case that expected free energy in its general formalism can be reduced to a reward function you have to have the information gained on top so what that means is that the rat in the maze has to first explore to become familiar with the maze before it becomes exploitative in an RL sense so I would actually challenge you that you're the Equality which you are selling which i think is a beautiful equality from the point of view the complete classroom is a slight misdirection because the whole point of the free energy minimizing expected free energy is it dissolves the exploration exploitation dilemma thank you I was just being lazy I mean of course like I you know I completely left out observations they were absent from those equations for a reason because I feel like they just getting away but that is certainly part of it as well right their information game was implicit what I was trying to do is just make a very specific link between reward and POW and default policy but it would have been included had I been more complete that's all I have to say I'm just saying I was being lazy there's sort of a high level thank you well that's it really or nothing else we're gonna hug it out in a second he did he did a sure this is gonna end in a hut so I'm gonna hold it to that at the end well maybe just to follow up on that some burning questions from the audience one of which asked explicitly does or can the free energy be principled capture none or l-systems and if not can it be considered unifying you can't see that here it's coming on this laptop so it does it capture non or l-systems is that a way you'd phrase it what you just described not capture non reinforcement learning systems I think it generalizes reinforcement learning so that's one way of carving up that sort of expected free energy is in terms of X reward which we both wrote down as a lot of probability or C for cost function price I didn't meet you did so that would certainly subsume all the machinery conceptual mathematical that you may want to bring to bear if you're using RL by repeat that's not the complete picture you need to equip that RL like imperative with an information game which is what I was referring to with the Lindley experiment the imperatives a good experimental design that you know what in the absence of any rewards there is still an imperative to go and get good data that is good in the sense of affording new information resolving uncertainty and I think that is exactly what you were talking about in terms of the good surprise you know it's a resolution of uncertainty afforded by epistemic affordance which mathematically would be the you know the the thing that you'd have to add to an RL cost function in order to get an expected free energy but did you mention inverse RL no you didn't I mean I imagine that's another so twist on the sociology which but I think to do that properly you're gonna have to put epistemic spurring the AIP specs back in the resolution of uncertainty as a fundamental part of the objective function sometimes I miss aliens sometimes known as novelty I call it epistemic affordance cuz I like that Giovanni Pazuzu so tell me just call it that I do good and so there's an well a question from Andre but maybe is it following up on some of these so there's questions on progress for this field and in terms of how do the models come about initially and how do we go about carefully driving process theories from these assumptions are somehow inferring the P and Q from data is that where we're going next then is that something the theory has helped us with or without narrow our investigations let me see the question yeah so this is the how do you choose your P's and Q's and theirs does not I don't have a I don't have a quick and ready response I can tell you what we do in practice which is we try to come up with sensible ones you know for example sparseness on natural images or things like that and we just see how and at the end of the day we kind of just see how they work right we see what sort of what sort of things would be predicted what kind of behavior is it consistent with behavior we often use very simple tasks when we talk about things like the Bayesian brain so we have very simple peas and very simple cues that are often drawn that are often the actual peas from our task right that's often how a Bayesian brain experiment goes you as an experiment or know what the pea is and you're really testing to see if the brain can figure that out too through experience as for queues you know if your optimal then your queue is just your put your just your posterior looking for causes of sub optimality is how is one of the ways in which we can identify the limitations of optimal inference and that's how we would then select our queues here's a simple experiment so this is my favorite experiment we talked about this one all the time you're trying to do auditory visual localization one possibility is that you're doing it optimally so you take into account uncertainty on both of them another possibilities that you don't and and so that would course so that you do means that on every trial your cue is a mean and a variance for both audition to audition reloj collation visualization and and if you're not going to do it in Bayesian way if we see that it's not coming from behavior the explanation would be that you're not tracking the variance on a trial by trial base you have some kind of fixed variance in mind ultimately I think that's how you address these questions right you build tasks for which you can do the inversion you know what the P is you look at the kinds of deviations people have humans monkeys whatever mice have from optimal behavior and try to come up with explanations for them by manipulating but by coming up with restrictions on Q that says nothing about the neural code so hope that your question really wasn't about that and then how did the models arise in the first place so do we just go back in early development of genetic priors or are there any insights that we can take from this principle for how they arise the generative models that you should showed were necessary for this theory but not necessarily for M or L based policy section so I mean the way that you'd normally are frame those questions in terms of hierarchical Bayes so yes certainly you can look at different time scales in terms of evolution your development experience dependent learning attentional control synaptic efficacy right down to faster actuations in responses to visual motion as essentially being reflections of a hierarchical genitive model where each level of the hierarchy affords empirical prize on on the level below and I guess in some sense could be looked at as doing a form of Bayesian model selection so if you have a hierarchical model questions like where did the prize come come from again just go away so they just come from the next level of and you normally end up with the evolution of the ugly as you intimated at the top level but that's not to get out you then compelled to model evolution as a process or natural selection as a process of Bayesian model selection and it all comes then down to define the relationship of the queue with the P but the Q is now the phenotype and the P is the environment and natural selection now becomes a process of the environment selecting those phenotypes that are fit for purpose in terms of them being able to model P the environment until you get to humanity when we start to design our own environments and then you get into the world of eco niche construction and free energy minimization underneath illogical which which you can go there if you want to so I'm not quite going to agree with that because what you what you've done is you've introduced yet a third P that's why I have a subscript for M right you said that P now there's a third P the P is the way the world actually works which is in my view I mean call this I know how many scientific anti-realists are there out there like that's that's inaccessible it's not a thing right the only thing that you've got is what you can store whether you're still unit in your brain what you can compute that's either what's in your brain or what's in your genes there is no P there is only a piece of em you'll cut me out as a radical skeptically oh yes I've done it before this is like when every once in a while someone says what's the most controversial position you've ever taken it's I'm not sure you guys exist I place I place no metaphysical significance on the latent variables of my models I promise not to use philosophical terms and I was violated I apologize so I'll be the structural realist then if that if that helps me so the position that I would be forced to take it with which probably is not what I'd want to if I was in a controversy and is that there is something real out there and it is the P and the Q and the P that the Q is predicated on the sort of implicit Charenton model which I agree entirely is not the real P it's not the generative process is is constrained by the real P out there felis I think it's a really interesting position because the whole sort of inferring from within your the skull when you just have access to data and everything everything that you do is based upon an inference about what's out there that you'll never ever actually directly connect with really does force you into a skeptical position philosophically the problem is that the free energy really does depend upon the goodness of those P's and Q's in the head in relation to something that's generating their sense of observation so you sort of force back into a realist position just to motivate the machinery so philosophically it's very uncomfortable for me but you are clearly comfortable being a skeptic a skeptical taking a skeptical position so that's nice there's a really nice question here and then will them go to Andre which is well theories developed differently in the context of the free energy framework if this isn't a theory no they'll go on with that it's a length okay sorry Mike and soapbox it's it's a language it's a way of talking about theories of brain function in my view it's a very useful one but again we can shoehorn because of the freedom to choose Pisa or P's and Q's we can chew horn just about any computational framework into it and so no it would not develop differently other than the fact that people might start talking about reward a little differently that's a possibility I think that's been talking about reward and default policy has been an exchangeable might be a good thing it might be helpful in some sense again it's certainly the case that we measure policies and we infer rewards and maybe there'd be some benefit to that I don't know so no one's made me a little strong I'm just wondering I mean in a sense you you intimated several ways that you could use the framework just to elaborate a whole bunch of process theory is predicated on different kinds of generative models but then you could go to as a neuroscientist you could go to the empirical data I'm one thing that's hit me in the past few years is a fundamental distinction between generative models based upon discrete space bases like Markov decision processes and hidden hidden Markov models and the like and those built around the assumption you're dealing with continuous states like Bayesian filters can't predictive coding and the like and just knowing that there are different sorts of judgment what was a carve in this way or that way I think and can put a lot of very useful constraints on the different hypotheses or process theories that you could then say well does it look as though the brains doing this kind of message passing or that kind of message passing or both and then we can get into the I'll grab the neural code what is what messages are actually being passed and how is it fit by physically represented so I guess I'm agreeing with you that the specification although the free energy principle has nothing to say about it when you start to commit to a process theory and you start to commit to the the generative model that will be apt for this experimental power you do actually narrow the hypothesis space about the biophysics that would be doing the you know the inference and thus the kinds of behaviors and neurophysiology that you would see from that and then you just base it model selection in the normal way to adjudicate thanks for your presentation so I think in both presentations we focused a lot on the needs and also on the Mystics and our physiologist mystics actually also that's not my full question I'll just say that we are then fine then so we have the physicists and the physiologist psychologists and so on on the other side but in between the Scruffy's we haven't discussed that very much and of course a substantial part of the conference are people from engineering and that are trying to use some of these principles to make better models so I was wondering if the either the free-energy principle or theory has informed that engineering very practical approach and if so how and the other question because we're in an interdisciplinary setting here it seems like we are in a perfect setting to explore the the confluence of those three groups together and I was wondering what you think we can actually learn from each other and what the interactions should be between the Scruffy's the needs and the Mystics so I'd be able to remember what those three things are so Scruffy's were engineers as I write needs were physicists what was the third one mystics mystics were philosophers and then where were we putting the physiologist with the mystics yeah I don't know how the free-energy framework particularly can bring these groups together or explicate their differences but I do see a general way forward that is related to like the Bayesian brain hypothesis which is that the role of the engineers is to enhance that's terrible is to enhance our computational space of models to give us more peas for which we can find queues and actually do inference so this is like saying we can use artificial intelligence to help us figure out maybe how real intelligence works physiologist obviously provide the constraints on what biology can do which also tells us something about behavioral you know the cues that are accessible I think to some extent and philosophers I'm sure there's one here all the ready philosophers here philosophers can tell us what the right words to use are do you want to go for that one too also someone has asked her car and minimize my free energy by speaking a bit louder okay sorry right this is very enjoyable by the way keep it coming we're trying to get be interested in your answer to be quite honest when you do electrophysiology you are well versed in the active inference and some of some of the maths so when you've got your electrophysiology cap on do you regard yourself as high-church theorist theorist or are you are you are missing I would have actually put you with with the Scruffy's I think electrophysiologists need to have a formal understanding of the of the system that they are trying to characterize and they are in the game I think of testing the process theories that we've been talking about doing the real heavy lifting but they need to be able to appeal to some theoretical framework in order to you know to provide mathematical rigor to their an hour season to that into their conclusions my experiences that they choose what works so that would that would come bring them into the Scruffy's camp and you're hearing the dialogue here as to why it's good to be a neat or perhaps you're not what do you think yeah I like being in the with the Scruffy's doing something very practical yeah but of course it has to be informed the the the we have an infinite space of possible experiment so we need to basically use a theory to inform what experiments we will do and even if a theory is conceptual only conceptual its you still need to have something to ground yourself in as a physiologist I was I was also wondering though just in terms of the pure computational space you know people from AI and so on has there been any progress in using free energy principles in that space and if so could you talk about them a little bit it's not my field but Mike's my limited knowledge of it is it's making more headway as one might imagine in things like neuro robotics so we talked about the construction real-world artifacts so the on the purely software side deep learning is the oil tanker that is moving forward and that's what everybody uses to be controversial I think that's something exactly in the wrong direction and it's obvious why well sorry it should be obvious why I say that not why it's in the wrong direction and it's all about simplicity it's always about carefully choosing the smallest amount of data in the algorithmically minimally complex way to get to the same reduction of uncertainty or increase in evidence so that means the challenge is using small amounts of data very carefully selected to solve the problem that's not what current sort of industry artificial intelligence is interesting it's going exactly the other direction exploiting the availability of very large amounts of data labels exploiting leveraging Moore's law and actually celebrating the computational complexity for me that's obscene because the computational complexity is part of the variation free energy it is fundamental this variational principle of least action so I haven't seen much movement in that and indeed we have been told by the couldn't great in charge of the money in this the you know some bright-eyed youngster like you or something like me is not going to change the direction of this oil tanker that's where it's going but in neurotics certainly is a non industrial scale where you actually have to make real things actively and to engage in interact with the world I see active inference making a much more simple elemental and fundamental difference you know just putting servers and playful reflexes inferences planning on top of existing robotics starting to think more about sort of interpersonal Dalek exchanges with robotics and the like I say that's probably industry wise making a difference from probably purely engineering and engineering perspective and just on the physics I've one question here from Roth Hafner at the University of Rochester who's asking does the free energy principle also apply to purely physical systems if so how is it different from Hamilton's principle of least action and if so what does it add and if it does not apply to purely physical systems wouldn't that make it falsifiable after all and just to mention there's been at least 10 or 15 people asking whether the theory is falsifiable so maybe you could but the falsifiable that's you know you religiously not use the word scheme you'd be very careful in the use of web frameworks by potters yes so the free energy principle the way this story was told here was concise almost at the point of being tie in cheek about you know what the underlying equations are what it would look like from the point of view of biological sciences but the back story is a version of Hamilton's principle is at least actually don't actually Hamilton's press release action it but it is a variational principle of least action it it certainly applies to physical systems in fact we'll go a little bit further it would say that physical systems inherit from the same variational principles that give rise to the very to the free energy principle so the idea is you've got this sort of variational principles based upon random dynamical systems when you have random fluctuations that dominate those you get quantum physics when you deal with ensembles you get statistical thermodynamics or more precisely stochastic thermodynamics when you take random fluctuations out of the game you get classical mechanics and then you get Lagrangian Newtonian mechanics and Hamilton's Hamiltonians come to the fore and when you put the Markov blanket which we haven't spoken about but if we frame this self-organization in terms of inference you then get the bayesian mechanics which is the free energy principle so it doesn't explain physics but it inherits from exactly the same maths that quantum physics and other physics explained again it's not falsifiable it's not there to be falsified it's not a theory that requires evidence it's just a mathematical truism clear well I've won for Jeff here so so I'm out cuz I was following up and which i think is an interesting question which you've touched on on ramstad a toss 20-19 work our generative models better conceived as encoded in the brain or enacted by agent and environment where does the model live if it has a subscript M it's in your brain I think that's my short answer it's it is part of the definition it is part of the definition of your computational energy is the set of available generative models right and then the other thing that's important is the restrictions on the set of posterior distributions that you're capable of computing and representing so brain of course as an extreme skeptic there is no there is nothing outside of that so I think that's why I have my answer that I'm happiest I'd agree it's the free it's the free energy gradients that drive the dynamics you need to define the generative model to define the gradient but it's the gradient itself that does the heavy lifting therefore strictly speaking there is no genitive modelled sub M it's just that ones which would be necessary to explain the gradient flows and the dynamic he's frowning we found we found a bunch of controversy sorry it sounds like you're saying you don't need to know you don't even even have a generative model as long as you have the gradients I think from the point of view of a physicists that's probably right yes but if you wanted to interpret what if I don't want to do gradient descent I like I like I like coordinates sent personally and so those gradients are a little different I think even with when you move to a coordinate descent so I don't mean to say that you can write down a scheme that will optimize something or self-organized your particular attracting set of states in the absence of generative model I'm just saying the genitive model is a mathematical construct that you will only find it expressed in the machinery doing the message passing the coordinate descent or the gradient descent in the brain itself so you'll see reflections be like cortical hierarchies in a hierarchical general jente model the factorization into wear and what into in terms of mean field approximation of latent causes of the sensory input but mathematically the genitive model in our self does not exist as a physical object is that still true when you're doing model comparison inside the brain because you have multiple generative models that are currently in sync because it seems to me in that situation you do need to evaluate the probability of the data given each of your models for the purpose of comparison which is that you would need them in some sense yeah I think it's true to sense in what one is me trying to observe your bread and trying to work out what generative model you are implicitly using sometimes said to in tailing the old brain entails a generative model to make it clear where there's no commitment to a physical instantiation or the genitive model it's just the the message passing is consistent with you using that guarantee model so that would be me as a neuroscientist looking at your brain and trying to infer your Geritol model I think from the point of view actually using a journey of model that's a slightly slightly which a simpler situation which I'm sure you can capture under your sceptical position it says it's an interesting way the question was framed you are we a generative model or are we a process a policy that can be described as if it was optimizing a genitive model defined by its priors since its sake it's a subtle question and it comes back and also irrelevant to the skeptic yes yeah you'd have to worry about it but it is consistent with your focus on defining phenotypes in terms of the things they do their policies so yeah you don't yeah thing is a policy it is not the genitive model that explains why you are doing that planning as inference there's a question here on Jeff's default policy and from Roberto Ghoulie at Columbia who says what's the reasoning behind choosing default over prior so who gave out you and how does it relate to the ability to potentially update it if it's a default right so you do update your pop you do update your default policy as soon as you receive sensory information which was the part that was left out of those particular slides which I've already apologized the reason why I don't use the phrase policy prior is because of the visceral reaction that I received from that man I mean that that litter that literally is the whole reason I mean so maybe so I think that maybe someone should hand Zaca mic and tell and ask him what what why why the visceral reaction Carl you put up on the on the screen Q of Pi which is a prior over policies and Jeff you put up on the screen P of s prime given s which is a default dynamics those are two different things and I don't know what the relationship is between them but they're different and our policy by itself is the pulse the probability of choosing actions given a belief state okay so if I'd simply put a prior on a that's already not a prior on a because I changed notation halfway through if I'd used PI and Q of PI instead of you I could I would have been allowed to call it a policy prior it was the way that you it's the meaning of what you were using the symbols were talking about the default dynamics of the states which is not what you do that's what you assume happens when you're just letting things go so that's equivalent to a reward yes so in practice and not in my example there is in fact a policy prior you got to ask somebody else for that one I don't know no because there is a queue of pie right sorry there's a P of pie and pie is empires a probability distribution over distributions I could take a vote from my point of view i I just flipped in planning as inference because it is to demystify all this sort of epistemic very instrumental value in the connection between our land and optimal Bayesian decision theory the expected free energy performance means just planning as inference that's all it is but it's just it incorporates both the normal optimum Bayesian design the Andreea as a little physiologist you are complying with those principles when you design your experiment you're working out what data should I go and sample solicit from my lab world you know to reduce my uncertainty over the different hypotheses that I am entertaining at the moment and I think the answer was that these frameworks provide the good hypothesis space that means you don't waste your time acquiring data that doesn't resolve at the uncertainty the queue of a PI's a posterior in fact so planning as inference is just how would you optimize your posterior beliefs about what to do next so PI is not a mapping between it's not a straight action policy as in RL optimal control theory where it's actually a mapping between I'm in this state and I have to do that so the active interest rate is much more general and deals with sequential policy optimization where there are multiple policies or sequences of actions in play and they're denoted by Q and as you say they have a posterior which means they have to have a prior interesting in the prior is though is the the softmax function of the expected free energy so from my point of view you can have them but possibly not in the sense that you were arguing about can I just so if active inference give us something on top of reinforcement length is there a class of tasks that would be solvable by one framework or Theory versus the other yes sequential process optimization in the case where you need to go in sample data that will change your beliefs so I noticed you use the word belief status a lot of confusion about that so a belief state is actually a state of belief that has all the sufficient statistics including the uncertainties and of course to encode completely would require you to have an infinite dimensional Markov decision process to do that so problems in which making that move resolves uncertainty about what would happen if I then did that so do I go to a restaurant I do I go to the do I make my own meal at home requires you first to resolve the uncertainty about whether you have the right ingredients in your cupboard or in your fridge so your policy has to incorporate your belief state your primaries at the point of acting knowing that your beliefs will change as time goes on so there is no strict RL scheme that can solve that problem is easier sometimes then reformulate mathematically RL could lead to solutions of new paradigms sorry formulation mathematical formulation has a modest value that's my comment so first of all I will sheepishly admit to being reviewer too but in my defense I don't think alive living I don't think alive as a territory but I had a I had a real question which goes back to falsifiability which I think a lot of people are interested in here and and let me just make sure I understood Jeff's proposed experimental strategy so you're thinking about defining some generative model imposing it on an experimental subject so that you can do you know what the optimal policy is the optimal posterior is and then looking for deviations and thinking and trying to examine to what extent you could explain those deviations in terms of some restriction on the posterior is that correct yes yeah so so that strikes me as not being a strategy for falsification because assuming that the way in which you pick yours you're looking for restrictions such that your that when you minimize free energy those restrict that that restriction will saw will satisfy the minimal free energy that's maybe the implicit in the way that you've set it up but maybe not it seemed to me that what you really want to be doing is impose the generative model separate or at least separately elicit people's assumptions about the generative model and try to then characterize their prior beliefs in terms of some restriction on the variational family before they've seen data right and then ask after they've seen data would the posterior within that restricted family reduced free energy and it doesn't note that it doesn't have to minimize free energy because so that kind of short-circuits the question about the algorithms because even if you just take a gradient step in that direction you're at least reducing it but but if the free energy ever went up if people ever chose a an approximate posterior that increased free energy that would be just I think decisive falsification of the free energy framework conditional on having pinned down all the other pieces well perhaps not like not enough samples is a reason why free energy can blip up well it depends how you mean when you're talking about samples so there could be a sample based variational Family but that then that would be part of your specification of the variational family that you do in the first part of the experiment or I think what you're saying is may be that people are approximating the free the expectation with a bunch of samples and they didn't take enough samples is that that's one pot so I mean in my experience there are two reasons why free energy goes up it's usually because that's the hardest thing to code up yeah is the evaluation of free energy for me personally so in other words it's a bug and the other cause is is is because I'm doing something that's time using some kind of stochastic method mmhmm right that's the only thing that ever causes it to go up in my experience but then again how would we evaluate that behave really like I don't know wait I just want to clarify your question yeah what you're actually falsifying in the silly example I gave is that the brain is optimally Bayesian right No yeah yes but well on the task right I could probably it's pretty I can it I guarantee you that there are some but in general right you know there's plenty of examples where the brain is not optimally Bayesian right that's right interesting question that's right those and those are the interesting examples right because that's where we can start like nailing down the question okay the falsifiable thing is is the brain Apollonia Bayesian know now we need an explanation there are a variety of different explanations all many of which take that would take two forms at least within this framework right wrong P wrong Q or severely limited Q and we have to explore those we can do model comparison to see which one is right completely within this framework in terms of right as in best fit to behavior and that's that best you can do so the thing is the funny thing about type of falsifiability is that is that to a large extent it's not like it this is not a falsifiable theory most theories of interest complicated theories are not really falsifiable in my opinion right it's really we would love to live in that we would love to live in I'm just gonna finish it it's okay trust me I'm just gonna interrupt you until I'm done I would love to live in like a nice world of falsifiable theories but at this point we don't have any right we have descriptions of behavior some are better than others and we use model comparison to say this is right this is wrong very interesting that happens a lot of times when we're looking at human behavior is that there isn't a single model that fits everybody it's incredibly rare people use a variety of different strategies to solve exactly the same problem often with very minimal differences in reward rates you know it's talked about specifics later it's not really important now so where does that leave us that leave us with the best description of behavior and again this is now the skeptic talking one more time last time I promised oh crap Isis gets doubt and what I was gonna say no that was okay yes no okay at the end of the day science in my opinion is about prediction and data compression and not about truth it never bothers me at all when there's a win up when it when if when a hypothesis is not falsifiable I only I really only care about those two things and prediction is really the only thing I can think and if I'm marginalizing I mean that's the automatic Occam's razor effective like doing Bayesian inference right yeah you got a bunch of models that are way over parameterize they're total they're not really useful so what do you do with them low posterior probability they're gone you're doing great prediction because you've focused on the models that actually work but my point was I guess less about falsification and more just about is there some distinctive prediction specifically of free energy you know that that is independent of the particular choice of P and Q and the the core prediction has to be that free energy has to go down right and so to fix so that so if it got you know if it if it doesn't it's because I you're using the wrong cue that's the sense in which is not falsifiable this that's it directly still on the board that SG is you paraphrase slightly oh I see SG easier good thanks more more on falsifiability because I I would you know if I can leave this debate and not spend the next several years testing a hypothesis that cannot be true or maybe it's trivially true I would like to avoid it so just to bring it back to some of the work that we've seen today you know the the the Scruffy's have over the past several years a bunch of neural networks they they take images and they make do some computation and they produce activity states and some of them are generative and some of them are not and one of the one of and what you can do with each one of these networks is take these activity States and regress them onto the brain and you get a prediction and right now one of the neural networks that gives us really accurate predictions reasonably accurate predictions is Alex needs to discriminate network and it's not generative so I don't know I mean that that's a hypothesis it could be true or it could be false and so it seems like one of the things that might be falsifiable here is the assumption of a that P of M lives in the brain because there are other computing things that are not generative models unless I'm just not thinking about Alex node as creatively as they need to I don't know if it's a question of thinking creatively if you were a scruffy you you wouldn't eat her I mean Scruffy's is not meant to be a derogatory term yeah I think the majority of ours are Scruffy's and but it certainly is easy to reconceive those d convolutional networks we saw in the last talk in terms of an amortized inversion of a generative model in fact there'd be a very graceful set of schemes that you could if he went to you to any commercial AI company the odd that could write down a variation altering code and then amortize that and you would end up with with the source of schemes and in a sense and I love that presentation because you know for me but what what was being done was essentially different forms of jointing models were being assumed and they were being for convenience amortize using current you know deep learning architectures and being put forward as working hypotheses for a process theory for you know sort of broad scale function anatomy heart anatomy in the brain and you know getting that right factorization I think underneath the shape and the form of the models and at what point of their divergent ease of dual-task face versus object these are exactly the structure learning problems that are presenting themselves when you don't know the form of generative model so I think it'd be quite easy to with the right rosetta stone start to talk the language that I would understand to describe that very set of results but you have to make the con you have to understand that all of these deep networks that you are just universal function approximate is what are they approximating the mappings that matter what are the mappings of matter they are mappings from data space to police inferences they're just amortized or summarizing learning how to infer how to categorize how to predict how to and all of that should at the end of the day be described as a variational for energy minimization thank you in the interest of time and for the next session I'm gonna ask our discussions if they're satisfied and if they have any closing statements satisfied I'm not satisfied yet I'm still you ever be satisfied I'm waiting for something very specific he wants he's hot okay they're gonna hug thank you all for your patience of your tenants and your great questions